

A Simplified Kinematic Strategy for Rod-Cone Docking Contact

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Make things simple but not simpler

-Albert Einstein

1. Introduction

Contact has been a hot topic for multibody system dynamics for a long period of time. To accomplish the simulation of such problem, some special techniques are required to handle the strong nonlinear behavior caused by the discontinuity of contact. In large-scale simulation, fine-scale contact computation takes a lot of resources. To improve the computation efficiency, for some rigid contact cases in rigid-flexible coupled systems, simplified models can be put into use. To this end, in this document we mainly discuss a virtual rigid contact formulation for simplification.

2. Kinematic Description of Floating Frame of Reference

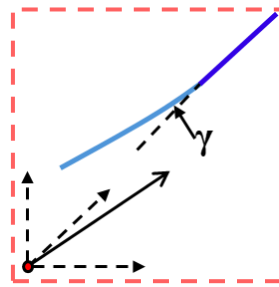


Fig. 1. Virtual rigid contact model: Shallow-blue curve stands for a flexible body, while the deep-blue one stands for the virtual rigid body

Still, we use space $\mathbb{Q} = \mathbb{R}^3 \times \text{SO}(3)$ to model the rigid body motion. The displacement $\mathbf{r}$ of one point on the flexible body reads

$$\mathbf{r} = \mathbf{p} + \mathbf{R}(\bar{\mathbf{x}}_0 + \bar{\mathbf{u}}_f(\bar{\mathbf{x}}_0)) \quad (1)$$

where $\mathbf{p}$ denotes the displacement of the reference point on the flexible body, $\mathbf{R}$ is rotation matrix of the reference point, $\bar{\mathbf{x}}_0$ is the static local position of points on the flexible body referred to the reference point, and $\bar{\mathbf{u}}_f(\bar{\mathbf{x}}_0)$ is the flexible deformation at point $\bar{\mathbf{x}}_0$ described in the reference frame $\mathbf{R}$.

As shown in Fig. 1, we assume that the flexible bodies are attached by massless rigid parts. And when contact happens, the forces are only computed on the rigid parts. In this way, we avoid the refined model of rigid body. That's why we call this method a "virtual" rigid contact model.

3. Simplified Rod-Cone Geometry Model for Virtual Rigid Body

Without the loss of generality, we take a simplified rod-cone geometry model in Fig. 2 for example. In this model, the contact can be formulated by only considering point-point contact when relative bias angle is small enough.

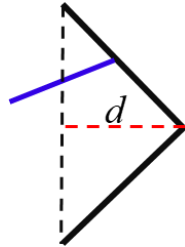


Fig. 2. Simplified rod-cone geometry model: Blue line stands for the virtual rigid rod and black cone stands for the virtual rigid cone

Based on Eq. (1), we further derive the position of rod tip,

$$\mathbf{r}_t = \mathbf{p}_t + \mathbf{R}(\bar{\mathbf{x}}_t + \bar{\mathbf{u}}_f(\bar{\mathbf{x}}_t)) \quad (2)$$

Still, $\bar{\mathbf{x}}_t$ stands for the static position in the reference coordinates. And it should be noticed that $\bar{\mathbf{u}}_f(\bar{\mathbf{x}}_t)$ is caused only by the flexible parts. By using the beam model in Fig. 1, $\bar{\mathbf{u}}_f(\bar{\mathbf{x}}_t)$ can be obtained. We use $\boldsymbol{\gamma} = [\gamma_x \ \gamma_y \ \gamma_z]^T$ to represent the small

deformation angles of beam. γ_x is the torsion angle while γ_y and γ_z are deflection angles in two directions respectively. If we make the contact model simpler, the rod is limited to a line, then the torsion effect can be ignored. Such that we have

$$\bar{\mathbf{u}}_f(\bar{\mathbf{x}}_t) = \bar{\mathbf{u}}_f(\bar{\mathbf{x}}_n) + \begin{bmatrix} 0 \cdot \gamma_x \\ d \cdot \gamma_y \\ d \cdot \gamma_z \end{bmatrix} = \bar{\mathbf{u}}_f(\bar{\mathbf{x}}_n) + \text{diag}(0, d, d) \boldsymbol{\gamma}_n = \bar{\mathbf{u}}_f(\bar{\mathbf{x}}_n) + \mathbf{D} \boldsymbol{\gamma}_n \quad (3)$$

where d is the length of rod and $\bar{\mathbf{x}}_n$ is the position of nearest beam node. If we consider a finite element model,

$$\boldsymbol{\gamma}_n = \mathbf{S}_\gamma(\bar{\mathbf{x}}_n) \mathbf{e}_n \quad (4)$$

Then Eq. (3) can be derived as

$$\bar{\mathbf{u}}_f(\bar{\mathbf{x}}_t) = \bar{\mathbf{u}}_f(\bar{\mathbf{x}}_n) + \mathbf{D} \mathbf{S}_\gamma(\bar{\mathbf{x}}_n) \mathbf{e}_n = [\mathbf{S}(\bar{\mathbf{x}}_n) + \mathbf{D} \mathbf{S}_\gamma(\bar{\mathbf{x}}_n)] \mathbf{e}_n = \mathbf{S}_r(\bar{\mathbf{x}}_n) \mathbf{e}_n \quad (5)$$

The direction of rod can also be quickly determined by make difference on the two-ends of corresponding nodes. We can then obtain the velocity of rod tip

$$\begin{aligned} \dot{\mathbf{r}}_t &= \dot{\mathbf{p}} + \dot{\mathbf{R}}(\bar{\mathbf{x}}_t + \bar{\mathbf{u}}_f) + \mathbf{R} \dot{\bar{\mathbf{u}}}_f \\ &= \begin{bmatrix} \mathbf{I} & -\mathbf{R}(\bar{\mathbf{x}}_t + \mathbf{S}_r(\bar{\mathbf{x}}_n) \mathbf{e}_n)^\wedge & \mathbf{R} \mathbf{S}_r(\bar{\mathbf{x}}_n) \end{bmatrix} \begin{bmatrix} \dot{\mathbf{p}} \\ \dot{\bar{\boldsymbol{\omega}}} \\ \dot{\mathbf{e}}_n \end{bmatrix} \end{aligned} \quad (6)$$

Also, if one wants to obtain the variation of $\mathbf{r}_t$, a similar operation in Eq. (6) can be used. Such that the generalized force, including the contact force, on the virtual rigid body can be expressed using

$$\delta W = \delta \mathbf{r}_t^T \mathbf{F}_{\text{ext}} = \delta \mathbf{q}_n^T \begin{bmatrix} \mathbf{I} & -\mathbf{R}(\bar{\mathbf{x}}_t + \mathbf{S}_r(\bar{\mathbf{x}}_n) \mathbf{e}_n)^\wedge & \mathbf{R} \mathbf{S}_r(\bar{\mathbf{x}}_n) \end{bmatrix}^T \mathbf{F}_{\text{ext}} \quad (7)$$

In this way, the contact on the virtual rigid body has transformed to the nearest finite element node.

Follow the same process, the position of arbitrary point on the cone can also be formulated. A slight difference lies in the position of cone that considers the contribution of torsion, which only requires the modification of matrix $\mathbf{D}$.

$$\bar{\mathbf{u}}_f(\bar{\mathbf{x}}_t) = \bar{\mathbf{u}}_f(\bar{\mathbf{x}}_n) + \begin{bmatrix} 0 \cdot \gamma_x \\ d \cdot \gamma_y - t(\bar{\mathbf{x}}_n) \cdot i_z(\bar{\mathbf{x}}_n) \gamma_x \\ d \cdot \gamma_z + t(\bar{\mathbf{x}}_n) \cdot i_y(\bar{\mathbf{x}}_n) \gamma_x \end{bmatrix} \quad (8)$$

where $t(\bar{\mathbf{x}}_n)$ is the width of cone at point $\bar{\mathbf{x}}_n$, $i_y(\bar{\mathbf{x}}_n)$ and $i_z(\bar{\mathbf{x}}_n)$ respectively stand for the local direction in cone at point $\bar{\mathbf{x}}_n$.

Finally, the contact forces are resolved using a penalty-based method or inequality-constraint-based method, this document does not further discuss.